# Oscillating Dimensional Trajectories on $E_8$ Lattices: A Recursive Framework Bridging Quantum Mechanics and Cosmological Scale

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October 24, 2026

**Abstract**

We introduce a dynamic theoretical model that treats spatial dimensionality not as a fixed background manifold, but as an oscillating discrete function $D(n) \in \{1, \dots, 8\}$ traversing the exceptional Lie algebra $E_8$ lattice structure. By defining a cyclic dimensional ladder operator modulated by powers of the golden ratio $\phi$, we demonstrate how high-dimensional geometric curvature collapses into lower-dimensional quantum state vectors on backward trajectory steps while expanding into inflationary cosmic geometry on forward steps. This deterministic feedback mechanism offers a unified geometric bridge between subatomic entanglement non-locality and cosmological horizon scaling.

## 1 Introduction

A central challenge in modern theoretical physics is reconciling the discrete, non-local behavior of quantum mechanics with the smooth, continuous curvature of cosmological spacetime. Standard higher-dimensional theories, such as string theory and M-theory, typically compactify extra dimensions on static Calabi-Yau manifolds.

In this work, we propose a dynamic alternative: an **Oscillating Dimensional Trajectory (ODT)**. Instead of fixed higher dimensions, physical state evolution occurs along a structured discrete clock $n \in \mathbb{N}$ that continuously climbs and descends the dimensional ladder from $1D$ to $8D$ and back, embedded directly within the $240$-root vector polytope of the $E_8$ lattice.

## 2 Mathematical Formulation

### 2.1 The Oscillating Dimensional Operator $D(n)$

Let $n \in \mathbb{N}_0$ represent the discrete evolution step. The active dimension $D(n)$ at step $n$ is defined by a periodic triangular wave operator modulo $14$:

$$D(n) = 8 - \Big|(n \bmod 14) - 7\Big| \tag{1}$$

This maps the step index $n$ to an endless dynamic trajectory:

$$D(n) \in \{1, 2, 3, 4, 5, 6, 7, 8, 7, 6, 5, 4, 3, 2\} \tag{2}$$

Under this mapping, lower dimensions on the reverse descent step ($n = 7 \to 13$) are mapped over the higher-dimensional metric states generated during the forward ascent step ($n = 0 \to 6$), enabling higher-dimensional topological information to be encoded into local low-dimensional space.

### 2.2 Golden Ratio ($\phi$) Modulation and $E_8$ Projection

The state vector $_n$ evolves according to a coupled projection operator:

$$_{n+1} = \mathcal{P}_{D(n)} \left( \phi^{(-1)^n \cdot D(n)} \cdot \hat{}_{E_8} \cdot _n \right) \tag{3}$$

where:

- $\hat{}_{E_8}$ is the lattice projection operator onto the $240$ roots of the $E_8$ exceptional Lie group.
- $\phi = \frac{1+\sqrt{5}}{2} \approx 1.6180339887\ldots$ is the golden ratio scaling factor.
- $\mathcal{P}_{D(n)}$ is the orthogonal projection slice mapping the 8D root lattice into the active subspace $D(n)$.

## 3 Physical Consequences

### 3.1 Quantum Scale Dynamics (Descent Phase)

During the reverse stepping phase ($D(n) : 8 \to 1$), the negative exponent in the golden ratio factor $\phi^{-D(n)}$ induces a logarithmic contraction of spatial coordinates.

Table 1: Trajectory Phases and Physical Manifestations

| Phase | Trajectory | Scale Factor | Physical Realization |
|---|---|---|---|
| Ascent | $1 \to 8$ | $\phi^{+D(n)}$ (Expanding) | Cosmic Inflation, Dark Energy |
| Apex | $D(n) = 8$ | $\phi^8 \approx 46.979$ | Holographic Horizon Density |
| Descent | $8 \to 1$ | $\phi^{-D(n)}$ (Contracting) | Quantum Non-locality, Entanglement |

Higher-dimensional proximity in $E_8$ is preserved during the descent, causing distant points in $3D$ Euclidean space to retain non-local topological connectivity. Quantum entanglement emerges naturally as a projection artifact of $8D$ adjacency folded into $3D$ projection slices.

### 3.2 Cosmological Dynamics (Ascent Phase)

During the forward stepping phase ($D(n) : 1 \to 8$), the positive exponent $\phi^{+D(n)}$ yields an exponential scaling of spatial bounds. This provides a self-similar, scale-invariant mechanism for cosmic inflation without requiring fine-tuned scalar potential fields.

## 4 Conclusion

The Oscillating Dimensional Trajectory framework provides a rigorous algebraic mechanism for unifying discrete subatomic behavior with cosmic scale expansion. By embedding spatial dimensional changes directly within a $\phi$-modulated $E_8$ lattice trajectory, the model opens new avenues for quantum gravity and quasicrystalline spacetime geometry.